A Privacy-Preserving Personal Social Graph for Measuring Social Connectedness

CSM500 MSc Project: Final Report

Victor Kironde

Student number 220241856

Submitted 12 October 2026

Word count: 6,889

Abstract

Loneliness and social isolation are recognised public-health concerns, yet people rarely have a clear view of their own relationships, and the tools that observe them are built for engagement rather than reflection. This project designed, built and evaluated *Social Health*, an Android application that turns everyday contact into a private, explainable measure of social connectedness. All data stay on the device in an encrypted, append-only event log. A transparent scoring model combines five weighted components (keeping to a chosen catch-up rhythm, phone-call metadata, relationship quality, balance across relationship types, and recent and frequent contact) and measures each relationship against the closeness the user hopes for, rather than assuming that closer is always better. The personal network is shown as a live ego graph whose layout, size, colour and grouping are driven by network metrics. Because no user study was possible, evaluation was specification-first: 23 properties and 17 scenario orderings written before the formulas, 284 automated tests, sensitivity and dominance analysis over 500 random networks, eight synthetic life case studies, and on-device performance and security checks. The model behaved as specified, no single component accounted for more than a third of the score's variance, and rendering stayed smooth on real hardware. The score's weights remain design assumptions that have not been validated against wellbeing outcomes.

Contents

1 Introduction **4**

1.1 Motivation 4

1.2 Problem statement and research question 4

1.3 Aims and objectives 4

1.4 Contributions 5

1.5 Report structure 5

2 Literature Review **6**

2.1 Social connection, loneliness and how they are measured 6

2.2 The structure of personal networks 6

2.3 Graph metrics for ego networks 6

2.4 Sensing social behaviour from mobile phones 7

2.5 Recency, frequency and composite scores 7

2.6 Existing apps, self-tracking and wellbeing risks 7

2.7 Visualising personal networks 7

2.8 Privacy, security and data protection 8

2.9 Gaps and what this project takes from the literature 8

3 Requirements, Users and Project Plan **9**

3.1 Target users and contexts of use 9

3.2 Functional requirements 9

3.3 Non-functional requirements 9

3.4 Potential harms and mitigations 10
3.5 Methodology and timeline 10

4 Design 12
4.1 Architecture: Model, View, Controller 12
4.2 Data model: an append-only event log 12
4.3 The scoring model 13
4.4 Graph model and metrics 14
4.5 Visual and interaction design 15
4.6 Privacy and security design 17

5 Implementation 19
5.1 Technology choices 19
5.2 Scoring engine 19
5.3 Storage, encryption and erasure 20
5.4 The live graph 20
5.5 Features built on the core 20
5.6 Notable engineering problems 21

6 Testing and Evaluation 22
6.1 Evaluation method and controlling for bias 22
6.2 Correctness: unit and property-based tests 22
6.3 Behaviour on synthetic networks 23
6.4 Case studies: eight synthetic lives 23
6.5 Sensitivity, ablation and component dominance 25
6.6 Performance on device 26
6.7 Security and privacy verification 27
6.8 Requirements traceability 27

7 Critical Reflection 28
7.1 Contribution against the research question 28
7.2 Limitations and threats to validity 28
7.3 Ethical reflection 28
7.4 Process reflection 28

8 Conclusion and Future Work 30

References 31

Appendix A: Scoring model parameters 37

Appendix B: Property and ordering catalogue 37

Appendix C: Threat model summary 38

Appendix D: Scenario details 38

1 Introduction

1.1 Motivation

Social connection is now treated as a determinant of health in its own right. The World Health Organization's Commission on Social Connection estimates that around one in six people worldwide experiences loneliness, and links it to roughly 871,000 deaths a year [1], [2]. Meta-analyses place the mortality risk of poor social connection alongside well-established risk factors such as smoking and physical inactivity [3]–[5]. Yet relationships rarely end by decision: they drift, and few people can see which are quietly fading.

Phones already hold much of that picture in call logs, calendars and message histories, but the social platforms that map people's relationships, such as Facebook and Instagram, are designed to maximise engagement rather than reflection [6], [7], process that data on remote servers, and in Facebook's case heavier use has been found to predict declines in wellbeing [8]. This project asks whether the same kind of picture can be built privately, on the user's own device, and presented in a way that informs rather than judges.

1.2 Problem statement and research question

Measuring a person's social connectedness from their own records raises three problems at once. The measure must be **technically sound**: it must weigh frequency, recency and type of contact sensibly, without letting one factor dominate. It must be **personal**: a relationship deliberately kept at a distance should not count against someone. And it must be **private and safe**: it concerns third parties who never agreed to be recorded, and a number about one's relationships can cause harm. The research question is therefore:

> *Can a privacy-preserving, on-device application turn everyday contact into an explainable measure of social connectedness that respects each user's own hopes for their relationships?*

1.3 Aims and objectives

1. **O1** Review the literature on social connection, personal networks, graph metrics, behavioural sensing, ranking by recency and frequency, visualisation and privacy, and derive requirements from it.
2. **O2** Specify an interpretable scoring model, with its expected behaviour stated as testable properties before any formula is written.
3. **O3** Design a local-first architecture and data model in which every score can be recomputed from recorded events.
4. **O4** Build an Android application with a live, metric-driven visualisation of the user's network.
5. **O5** Protect the data through encryption, minimal permissions, erasure and a documented threat model.
6. **O6** Evaluate correctness, robustness, performance and security without relying on the author's judgement alone.

1.4 Contributions

The project contributes an interpretable five-component connectedness model with 23 stated properties; per-relationship *ideals* that measure each relationship against the closeness the user hopes for; an event-sourced, local-first data model; a live, metric-driven ego-graph visualisation; and a specification-first evaluation method suited to a project without test users.

1.5 Report structure

Chapter 2 reviews the literature; Chapter 3 sets out requirements, users and harms; Chapters 4 and 5 present the design and implementation; Chapter 6 evaluates the system; Chapters 7 and 8 reflect and conclude.

2 Literature Review

2.1 Social connection, loneliness and how they are measured

Social connection is usually described along three dimensions: **structural** (network size, contact frequency, variety of roles), **functional** (the support relationships provide) and **quality** (how positive or strained they are) [9]. Loneliness, the subjective feeling that one's relationships fall short of what one wants, is distinct from isolation, the objective lack of contact: a person can be lonely in company or content alone [10]. Validated instruments such as the UCLA Loneliness Scale and its three-item short form, the De Jong Gierveld scale and the Lubben Social Network Scale measure these constructs through self-report [11]–[15]. Two implications follow for this project. First, a tool that observes contact estimates the structural and, partly, the quality dimensions; it cannot measure loneliness and should not claim to. Second, reviews of interventions find small and varied effects [16], so a short project should evaluate whether its system behaves as specified rather than promise wellbeing outcomes.

2.2 The structure of personal networks

Research on personal (ego-centred) networks describes one person, the *ego*, their contacts, the *alters*, and the ties among those contacts [17]. Dunbar's work suggests that people maintain roughly 150 meaningful relationships, arranged in layers of about 5, 15, 50 and 150, with closeness and contact frequency falling at each step outwards [18]–[20]. Relationships cost time to build [21] and decay without contact, but not uniformly: emotional closeness to friends falls faster than to kin when contact stops [22], [23]. Granovetter defined tie strength as a combination of time, emotional intensity, intimacy and reciprocal services [24], and later work found closeness a better indicator of strength than contact frequency, which overstates ties with neighbours and colleagues [25]. A score built only on frequency would therefore misjudge the relationships people value most. This shaped three design choices: separating the user's judgement (importance, quality) from behaviour (contact); letting the expected contact rhythm differ per relationship; and making past contact decay rather than accumulate.

2.3 Graph metrics for ego networks

Standard network measures describe an ego network compactly. **Degree** and **strength** count and sum a node's ties [26]; **density** is the share of possible ties among alters that exist; **local clustering** shows whether a person's contacts know each other [27]; **betweenness** measures how often a node lies on shortest paths [28], [29]. Burt's theory of structural holes values people who bridge otherwise separate groups [30]; for an ego network his redundancy measure reduces to an **effective size** of $n - 2t/n$, where n is the number of alters and t the number of ties among them [31]. Communities can be found by maximising modularity, efficiently with the Louvain method [32], [33]. Ties among alters can also be inferred from behaviour: in one well-known study, phone data alone identified most self-reported friendships [34]. These metrics are usually applied to complete networks gathered by researchers; applied to a self-recorded network they describe only what the user logged, and should be presented that way.

2.4 Sensing social behaviour from mobile phones

Phones record a great deal about social behaviour: call and message logs, Bluetooth proximity, location and app use. Studies have used such data to infer friendships and tie strength, estimate loneliness and social isolation, and detect depressive symptoms, with promising but mixed accuracy [34]–[38]. Recent work on recognising loneliness from smartphone sensing reports moderate classification performance and warns about small, unrepresentative samples [39]–[41]. These studies show that call metadata carries real signal about relationships, but also that passive sensing raises consent and privacy concerns, can be biased by how individuals use their phones [36], [42], [43], and is restricted by platforms: Google Play limits access to the call log to a narrow set of apps [44]. A design that uses phone metadata should therefore make it optional, minimal and weighted, never the only source of evidence.

2.5 Recency, frequency and composite scores

Ranking by how recently and how often something happened is common in software. Query suggestion and autocomplete systems rank candidates by counting past uses with time-decayed weights [45]–[47], and web browsers rank history the same way, so that a site visited often last week outranks one visited once a year ago [48]. Exponential decay, where each event's weight halves every fixed period, is the usual form, because it is simple, has one interpretable parameter and can be updated incrementally [49]. Combining several such measures into one number is the domain of composite indicators. Guidance on constructing them stresses transparent weighting, normalisation, sensitivity analysis and checks that no single component dominates, and warns that a composite can hide weaknesses behind strengths [50], [51]. These practices were adopted directly in the evaluation.

2.6 Existing apps, self-tracking and wellbeing risks

Personal relationship managers remind users to keep in touch at set intervals [52], while platforms such as Facebook surface connections to drive engagement. Their effect on wellbeing depends on use: composed, one-to-one communication with close ties is associated with improvement, broadcasting and one-click feedback are not [53], and passive scrolling tends to undermine wellbeing [7]. Evidence that digital tools reduce loneliness is weak: a meta-analysis for older adults found no significant reduction at three, four or six months [54]. A recent trial of a social-connectivity app with 99 older adults found lower depression but no change in loneliness after four weeks, and its authors call for work on the risks of reflection-based apps [55]. Personal-informatics research describes self-tracking as moving from preparation and collection to reflection and action [56], [57], and warns of fixation, anxiety and feeling judged by one's data [58], [59]. A relationship tool must therefore inform without prescribing, avoid gamification, and let users step back from the number.

2.7 Visualising personal networks

Ego networks are commonly drawn as concentric circles with the ego at the centre, a convention used in network-elicitation tools in which closer rings mean closer relationships [60], [61]. Force-directed layouts place connected nodes near each other and are widely used for general graphs [62], [63]; combining radial constraints with forces keeps both the ego's view and the structure among alters. Visual-encoding research ranks position and length above area and colour for precise comparison, so the most important variable

should be shown by position [64], [65]. Set and region overlays, drawn as shaded areas around members, help show group membership, but overlapping regions can be misread, which argues for consistent shapes and restrained colour [66], [67].

2.8 Privacy, security and data protection

Relationship data are personal data about the user and about people who have not consented to being recorded. Ratings of a relationship and notes about it are sensitive even when they are not special-category data in law. The UK GDPR's principles of data minimisation, purpose limitation and storage limitation, and the rights to access and erasure, apply to any service that processes them [68], [69], although purely personal use on one's own device falls within the household exemption [70]. Privacy by design favours local processing, encryption at rest and in transit, and collecting only what is needed [71], [72]. On mobile devices, platform guidance recommends hardware-backed key storage, disabling cloud backup for sensitive data and requesting permissions only when they are needed [73]–[75].

2.9 Gaps and what this project takes from the literature

The review reveals a gap: research on sensing and scoring social behaviour is mostly server-based, population-level and aimed at prediction, while consumer tools are engagement-driven and rarely transparent. Little work offers an individual a private, explainable and personally calibrated view of their own network. From the literature the project takes the following principles, which become requirements in Chapter 3: measure structure, not loneliness; separate judgement from behaviour; decay contact over time with diminishing returns; weight phone metadata but never rely on it; build any composite score transparently and test it for dominance; use network metrics to describe circles and bridges; inform rather than prescribe; and keep everything on the device.

3 Requirements, Users and Project Plan

3.1 Target users and contexts of use

The intended users are adults who want to reflect on their relationships, particularly at times of transition, when networks change most: moving to a new city, starting university, a demanding job, becoming a parent, or retirement [76]. Eight synthetic "lives" were written to represent these contexts, from a doctor who sees many unnamed patients to an older woman in a care home (Section 6.4). The tool is **not** intended as a clinical instrument, for people in acute crisis, or for monitoring other people. Because a number about one's relationships can be distressing, the design treats wellbeing as a requirement rather than an afterthought.

Two contrasting users illustrate the range. A doctor who sees dozens of patients a day needs work contact counted without naming every patient or masking a fading friendship. A widowed care-home resident needs a death not treated as neglect, family who call rather than visit to count, and a calm tone. Both shaped the requirements below: groups with unnamed attendees, an "in memoriam" status, call metadata as a separate component, and descriptive wording.

3.2 Functional requirements

Table 1 lists what the system must do, each traceable to a gap in Section 2.9 and verified in Section 6.8.

Table 1. Functional requirements.

ID	Requirement
FR1	Add, edit and remove people and groups, with relationship type, importance (1–10), quality and status.
FR2	Log a catch-up with one or more people or groups: when, how (in person, video, voice, messages) and an optional private note.
FR3	Set an expected catch-up rhythm and an ideal closeness per relationship, or follow defaults.
FR4	Compute an overall Social Health score and a score per relationship, explained by component.
FR5	Show the network as an interactive graph with filters, groups and inferred social circles.
FR6	Show how the score and relationships have changed over time.
FR7	Optionally use phone-call metadata and import names from contacts.
FR8	Plan meetups, once or repeating, and confirm whether they happened.
FR9	Export all data (encrypted) and a printable report, and erase a person or everything.

3.3 Non-functional requirements

Table 2 lists the qualities the system must have; privacy and wellbeing are treated as requirements, not features.

Table 2. Non-functional requirements.

ID	Requirement
NFR1 Privacy	No server or account; data never leave the device unless the user exports them; minimal permissions, requested only when needed.
NFR2 Security	Encryption at rest with a hardware-backed key; optional app lock; encrypted export; no cloud backup.
NFR3 Explainability	Every score traceable to its components; parameters documented and versioned.
NFR4 Robustness	No single component dominates; behaviour stated as properties and tested automatically.
NFR5 Performance	Smooth graph interaction (under 1% dropped frames) for at least 150 people on a current phone.
NFR6 Wellbeing	Descriptive, non-judgemental wording; no streaks or rankings of people; the score can be hidden.

3.4 Potential harms and mitigations

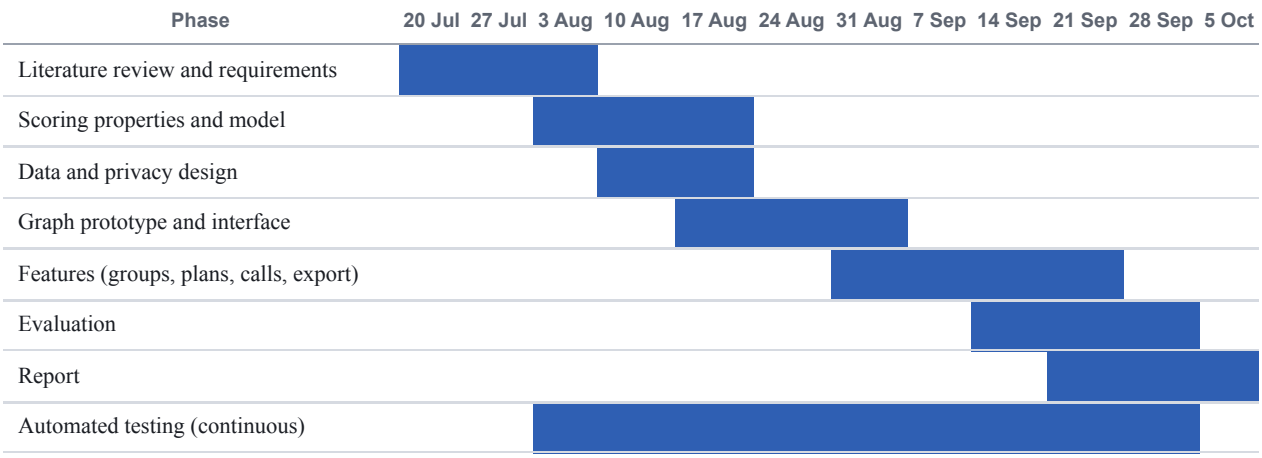
Table 3. Potential harms and how the design responds.

Harm	Mitigation
Fixation on the number, or reading it as a judgement of worth	Score shown 1–10 with gentle closeness labels; a persistent statement that it is for reflection only; option to hide it entirely.
Feeling rejected when someone close appears distant	Labels describe closeness, never fault; the score reflects only what the user logged; an ideal can be lowered.
Pressure to contact harmful people	A relationship rated harmful is excluded from the score, so contact never raises it.
Bereavement counting as neglect	An "in memoriam" status removes the relationship from the score with care.
Exposure of third parties	Local storage, encryption, erasure, anonymised report option (Section 4.6).

3.5 Methodology and timeline

Development was iterative and test-led. The scoring engine was written as a pure library, and its expected behaviour was specified as properties and scenario orderings in the first commit, before any formula existed. The interface was built in short cycles, each ending with a build tested on a physical device or emulator. Figure 1 shows the planned timeline; Section 7.4 reflects on how the work departed from it.

Figure 1. Planned timeline, 20 July to 12 October 2026, by week starting.



4 Design

4.1 Architecture: Model, View, Controller

The system follows the Model–View–Controller pattern (Figure 2) [77], with each part independently testable.

- **Model.** A pure TypeScript scoring engine and the event log it reads. The engine has no knowledge of storage or the interface: given a log, a configuration and a date, it returns the score, its components and a contribution for every relationship. A separate synthetic-data package builds reproducible test networks.
- **View.** React Native screens and a GPU-drawn graph. Every page except the graph opens as a pane over it, so the network stays in view while the user edits it.
- **Controller.** A store that validates input, turns user actions into events, persists them in the encrypted database, keeps derived state (groups, plans, graph positions) and synchronises optional call-log data.

The direction of dependency is strictly inward: the view depends on the controller and model, the controller on the model, and the model on nothing. This is what allows the same engine to drive the live score, trends, filtered scores, the PDF report and the evaluation scripts.

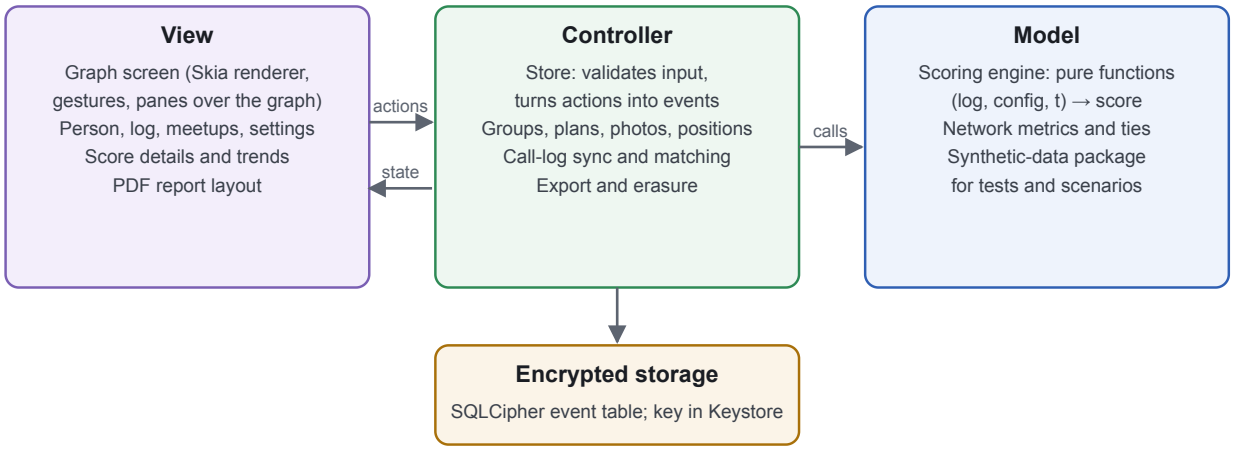


Figure 2. Architecture as Model, View and Controller; the model depends on nothing else.

4.2 Data model: an append-only event log

Everything the user records is stored as an immutable event: a person or group is added, a relationship attribute changes (type, importance, quality, rhythm, ideal, status, "lives or works together"), a catch-up happens, a call is imported, a period is paused. The current state of any relationship is the fold of its events up to a given time. This event-sourcing approach has three benefits [78]. The score at any past date is recomputed exactly from the events before it, so trends need no stored history. Changing a weight re-derives the whole history consistently. And every result can be traced to the records that produced it (Figure 3).

Groups (their membership) and plans (scheduled meetups) are kept outside the scoring log as current state. A group is also a relationship in its own right: a catch-up with "Book club" counts once for the group and once for each named person present, while people not named are stored only as a count. A planned meetup becomes a catch-up only when the user confirms it. Erasure is the one exception to immutability: deleting a person removes or rewrites every event that mentions them.

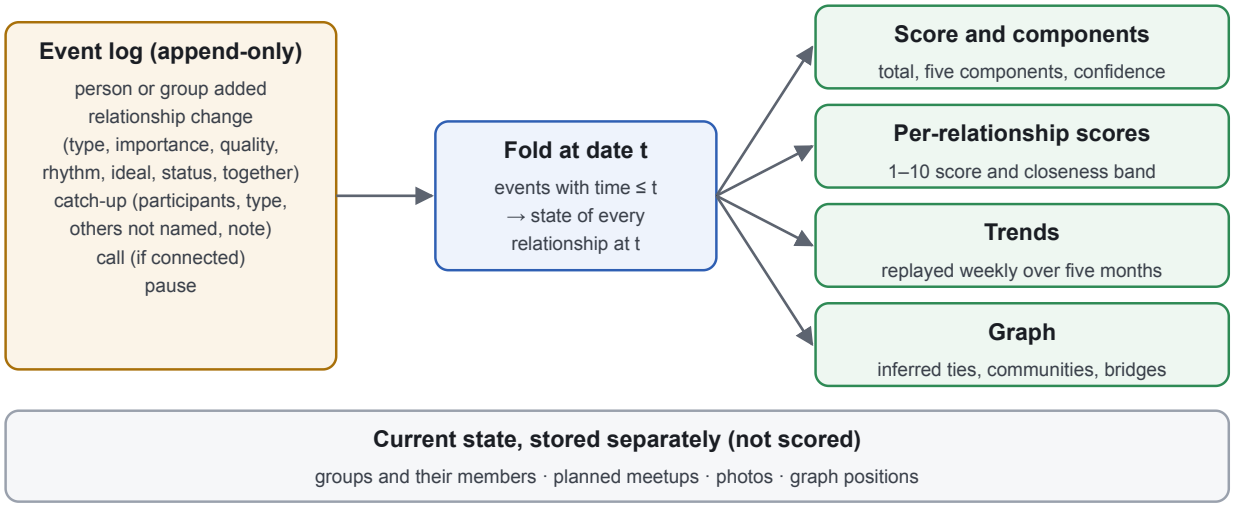


Figure 3. The event log is folded at any date into relationship state, from which scores, trends and the graph are computed.

4.3 The scoring model

For each relationship i, contact is turned into a **level** that rises with each catch-up and decays over time. Each catch-up x contributes according to its type (in person 1, video 0.7, voice 0.5, messages 0.2) and the relationship's quality multiplier m_i, halving every expected interval I_i:

$$E_i = \Sigma_x \, w(\text{type}_x) \cdot m_i \cdot 0.5^{\text{age}(x)/I_i}, \qquad L_i = 1 - \mathrm{e}^{-E_i} \tag{1}$$

The saturating form gives diminishing returns, so repeated contact with one person cannot make up for neglecting others. The expected interval comes from a rhythm the user chooses ("at least once a month" by default, or a custom frequency such as "at least twice a week"). For someone the user lives or works with, time apart counts a tenth as much, because they meet without logging it. Five components (Table 4), each between 0 and 1, are then averaged with importance weights (1–10):

Table 4. Score components and weights.

Component	Weight	Definition
Catch-up rhythm	0.25	Importance-weighted mean contact level L_i.
Calls	0.25	The same construction over imported calls (answered calls of a minute or more count fully, shorter ones 0.3, missed calls not at all); left out if the call log is not connected.
Quality and support	0.10	Mean of the user's latest quality and support ratings.
Balance across types	0.10	Mean over relationship types of the square root of each type's mean level: rewards balance across the types the user has, not having more types.

Recent and frequent contact	0.10	All contact with a 30-day half-life, relative to steady contact at the relationship's rhythm, capped at 1.

$$\text{Score} = 100 \times \Sigma_c\, w_c \cdot C_c / \Sigma_c\, w_c \quad \text{over the components } c \text{ that are defined} \tag{2}$$

The weights follow the proposal feedback, which suggested a 0.25 weight for phone metadata and a 0.10 weight for recency and frequency. A weighted arithmetic mean was chosen over a geometric one for interpretability [50]: each component is shown beside the total, so a strong component cannot silently hide a weak one.

Ideals. The standard model assumes every relationship should be as close as possible. That is not always true: a person co-parenting after a separation may want a cordial, distant relationship. Each relationship therefore has an ideal g from 1 to 10 (default 10), or "don't mind". With target $t = (g - 1)/9$, each contact-based value x is replaced by its fit to the target:

$$\text{fit}(x, g) = 1 - |x - t| / \max(t, 1 - t) \tag{3}$$

With $g = 10$, $\text{fit}(x) = x$, so the original model is the special case and existing scores are unchanged until an ideal is lowered. "Don't mind" removes the relationship from the score. A relationship rated harmful, archived or marked in memoriam is likewise excluded, so contact with it can never raise the score.

Display. The 0–100 total is shown as a whole number from 1 to 10 inside a coloured ring. Each relationship also gets its own 1–10 score and a descriptive closeness band (Close, Connected, Drifting, Distant, Strained, Lost touch), set by the weaker of its quality rating and its rhythm (Figure 4). A confidence measure marks scores built on sparse logging as uncertain rather than as a decline.

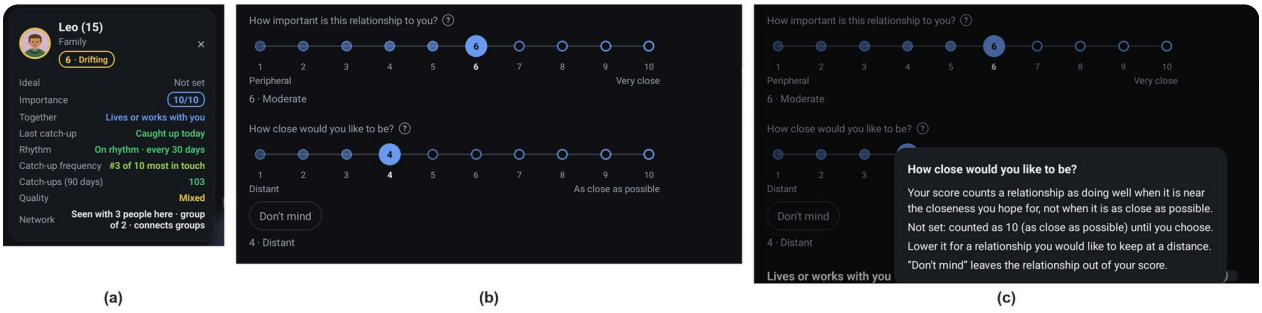


Figure 4. One relationship: (a) the stats card, with colour-coded values; (b) the importance and ideal scales; (c) a help tip.

4.4 Graph model and metrics

Ties among alters are not asked of the user; they are inferred from catch-ups with two or more people. Each shared catch-up adds $0.5^{\text{age}/90\ \text{days}} / (k - 1)$ to every pair among its k participants, a discount borrowed from co-authorship networks [79] so that one large gathering does not tie everyone strongly. On this weighted graph the system computes density, effective size, clustering, betweenness and Louvain communities, which drive the visualisation: communities shape clustering, people whose ties span communities get an extra outer ring, and groups are linked to their members and can be collapsed.

4.5 Visual and interaction design

Each visual channel carries one variable, chosen by how accurately it is perceived (Figure 5):

- **Position:** distance from the centre shows importance. A force simulation pulls each node towards a ring set by its importance and towards the sector of its relationship type, while collisions keep nodes apart.
- **Size:** five discrete sizes show how recently and often the user is in touch, ranked across the network.
- **Colour:** the ring and the line to the centre show closeness, from green (close) through amber to red (strained), with grey for lost touch; lost-touch nodes also fade and sit further out.
- **Regions:** each relationship type is a softly shaded rounded rectangle in its own colour, kept clear of the closeness colours so a region is never read as a rating; groups are drawn as rounded squares so they are never confused with regions.
- **Motion:** anyone in touch within the last week pulses gently.

People and whole regions can be dragged and stay where they are put, but cannot be pushed into an unrelated region. The score, its details, the filters and the help key all open as see-through panes over the graph, so the effect of a filter is visible as it is changed (Figure 6).

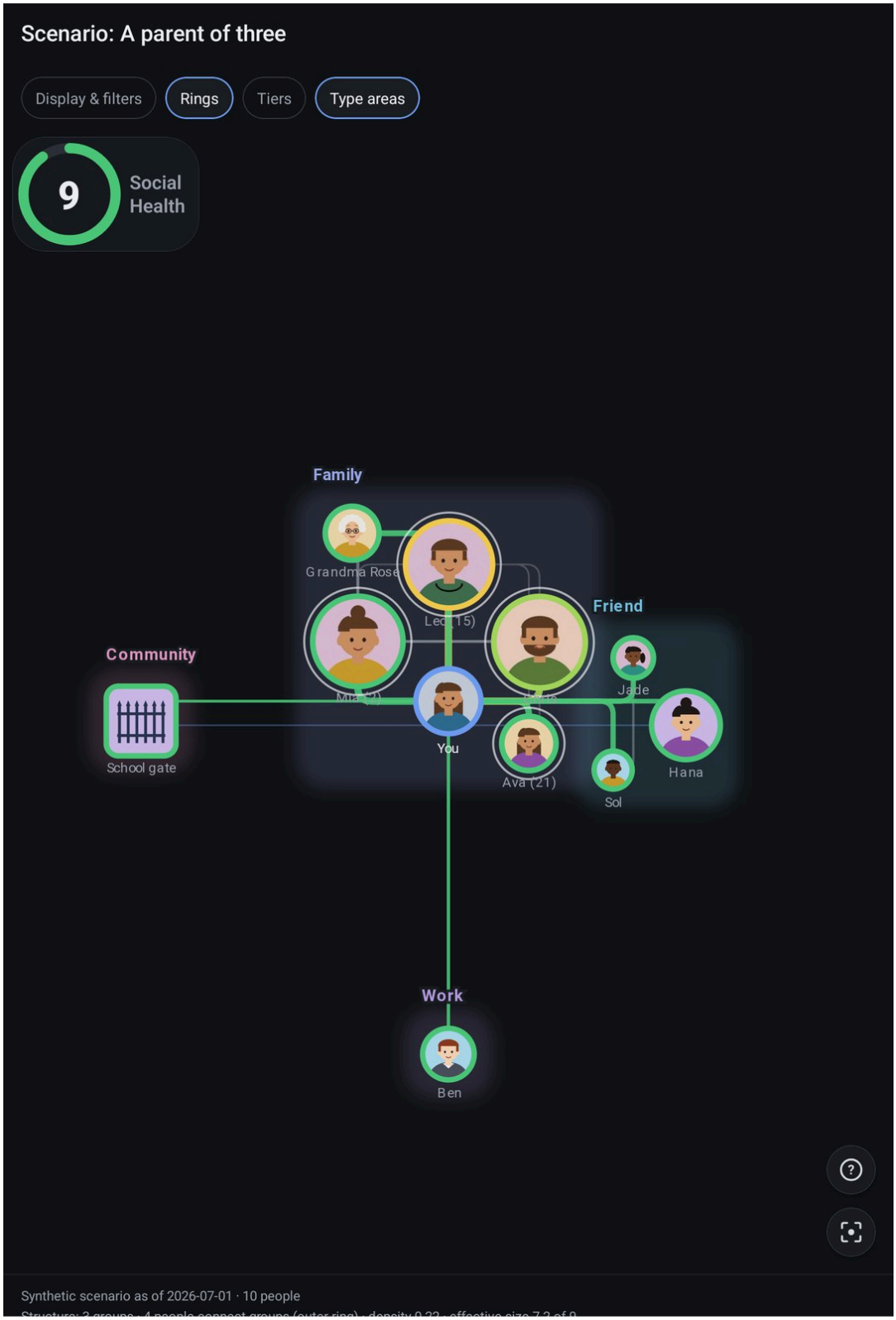


Figure 5. The network graph for the "parent of three" scenario.

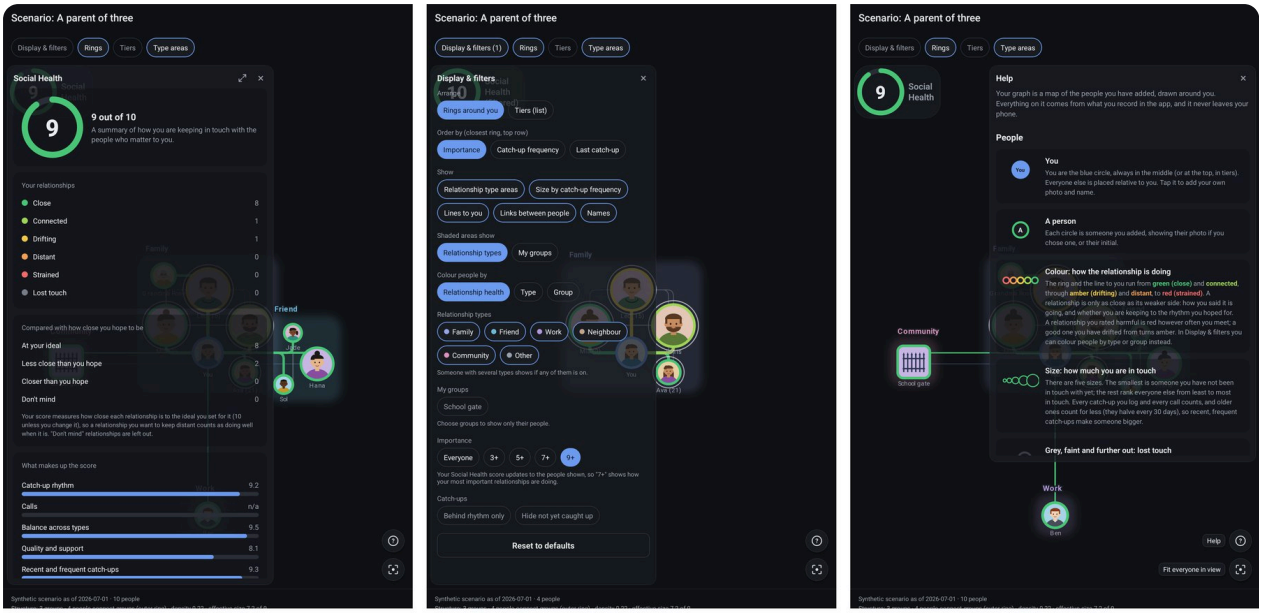


Figure 6. Panes over the graph: (a) score details; (b) display and filters, here "Importance 9+"; (c) help.

4.6 Privacy and security design

A STRIDE-style threat model [80] identified the assets (names, ratings, contact history, the database key, exports), the actors (someone holding the unlocked phone, a thief, other apps, cloud backup, export recipients and the recorded people themselves) and the threats. The main mitigations are:

- no server, account or network request: data stay on the device;
- a SQLCipher-encrypted database [81] whose random 256-bit key is held in the Android Keystore [73];
- an app lock using the phone's biometrics or PIN, on by default;
- cloud backup disabled;
- encrypted export with AES-256-GCM and a PBKDF2-derived key (600,000 iterations) [82]–[84]; an unencrypted export only behind a warning;
- contacts and call-log permissions requested only when the user turns those features on; the call log is read without phone numbers, and calls are matched to people by name on the device;
- erasure of a person from every record, and "delete everything" by destroying the database and its key;
- a printable report with an optional Confidential watermark and an option to anonymise people (Figure 7).

Tying the key to biometric authentication would make the app lock cryptographic, but such keys are permanently invalidated when a new fingerprint is enrolled [85], silently destroying the data; the lock therefore guards the screen, and encryption guards the storage.

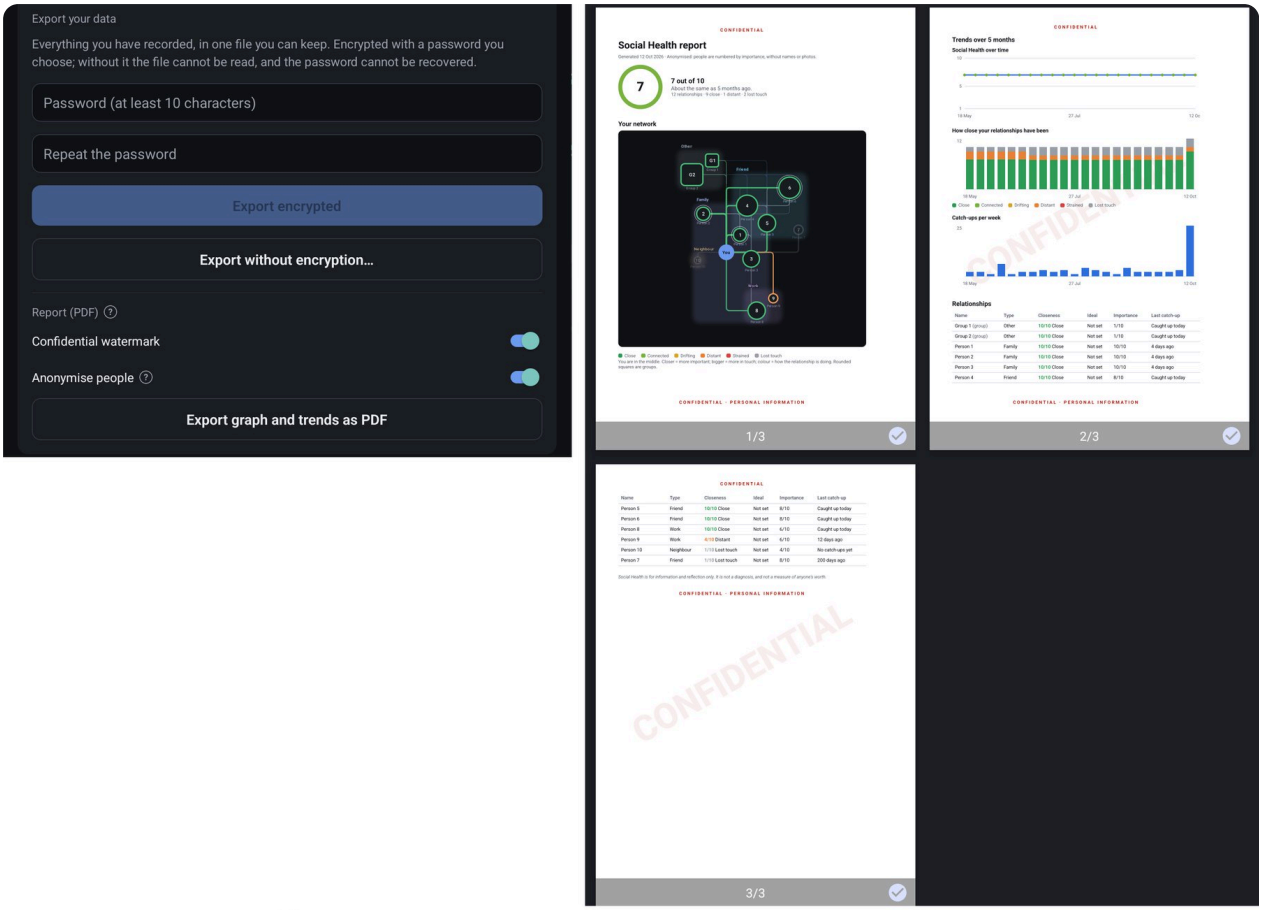


Figure 7. The printable report: (a) its options; (b) an anonymised preview.

5 Implementation

5.1 Technology choices

The application is built with React Native and Expo in TypeScript [86]–[88], so that one typed codebase covers the engine, the interface and the tests, and the engine can run unchanged in Node for evaluation. Table 5 lists the main technologies and why each was chosen.

Table 5. Main technologies.

Technology	Use and reason
React Native, Expo, Expo Router	Cross-platform native interface; file-based navigation; managed native builds.
TypeScript	Static types shared by engine, app and tests.
React Native Skia [89], [90]	GPU drawing of the graph; much faster than SVG for hundreds of shapes.
Reanimated, Gesture Handler [91], [92]	Animations and gestures on the UI thread, independent of JavaScript load.
d3-force [63], [93]	Well-tested force simulation for the layout.
expo-sqlite with SQLCipher [81], [94]	Encrypted local database.
Expo SecureStore, LocalAuthentication	Keystore-backed key storage; app lock.
react-native-quick-crypto [95]	Native PBKDF2 for export encryption.
expo-print, expo-sharing	PDF report generation and sharing.
Vitest, fast-check [96], [97]	Unit and property-based testing.

5.2 Scoring engine

The engine is a set of pure functions over the event log, with every parameter in one configuration object and every result carrying the model version (0.4.0). Trends call it at past dates, a filtered score calls it on a restricted log, and user preferences are applied by deriving a new configuration.

The engine was developed in four stages. First, the properties and orderings were written as executable tests against a stub that returned zero, so all of them failed. Second, the per-relationship quantities (levels, rhythm, call levels and recent contact) were implemented one at a time until the properties that depended on them passed. Third, the components and weighted total were added, and the sensitivity script was written to report how each one moved the scenario rankings. Fourth, later model versions were introduced only behind new properties: P17 was written before ideals were implemented, guaranteeing that the default ideal left every existing score unchanged, and P22–P23 preceded the "lives or works together" decay. Each version bump was therefore a controlled change whose effect could be read directly from which tests failed.

5.3 Storage, encryption and erasure

Events are rows in one SQLCipher table with a versioned schema and forward-only migrations. The random key is generated on first launch and kept in the Keystore, and secure deletion overwrites freed pages. Erasing a person deletes their events, rewrites shared catch-ups and removes them from groups, plans, photos and saved positions in one transaction.

5.4 The live graph

A custom Skia renderer records drawing commands as pictures on the UI thread. The force simulation advances one step per frame on the JavaScript thread and publishes positions to a shared value; once settled, positions are saved so the next launch starts where the user left off. A drag that starts on a person moves that person, one inside a region moves the whole region, and either stops at the edge of an unrelated region. For the PDF report the same pictures are replayed off screen, so the exported graph matches the screen.

The simulation is seeded, so the same network always produces the same layout, and a remembered layout is relaxed only when the ties or importance it was computed for change. During a drag the simulation runs cool, so others make room without the whole graph rearranging.

5.5 Features built on the core

Most features are thin layers over the engine and store. **Groups** (Figure 8a) gain members automatically when someone is named at a group catch-up. **Planned meetups** (Figure 8b) generate dates and ask afterwards whether each happened. **Call-log matching** reads only name, time, duration and type, discarding anything ambiguous. **Trends** replay the engine weekly, and the **PDF report** is rendered as HTML and printed to A4.

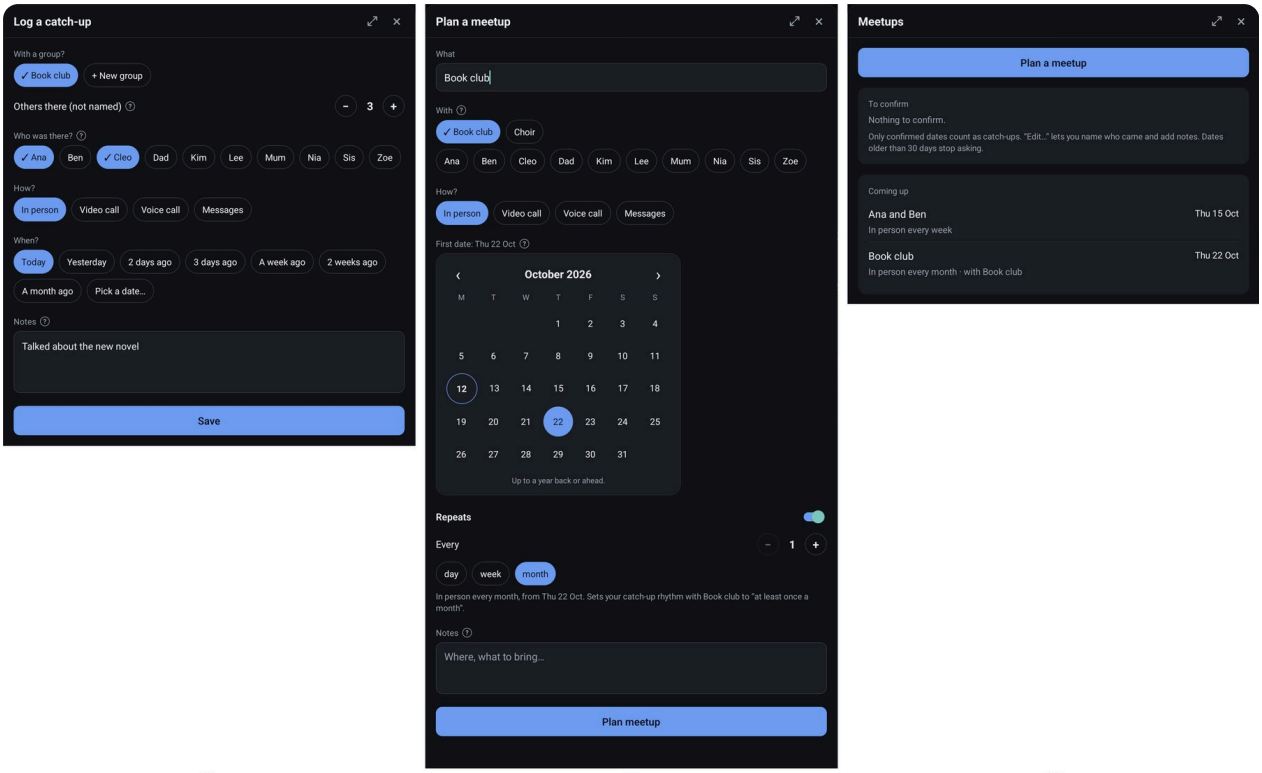


Figure 8. (a) Logging a group catch-up; (b) planning a repeating meetup; (c) upcoming meetups.

5.6 Notable engineering problems

- **Key derivation cost.** A pure JavaScript PBKDF2 took 81.5 s for 310,000 iterations on the phone, because the Hermes engine has no just-in-time compiler [98]. Moving it to native OpenSSL allowed the recommended 600,000 iterations in a fraction of the time, with identical keys verified against a known answer.
- **Drawing performance.** Drawing each link as a stroked path caused up to a third of frames to drop with 250 links; drawing all link segments in one call fixed it.
- **Disappearing graph.** When zoomed in, Skia judged recorded pictures to be off screen because they lay at negative coordinates; recording with an offset fixed it.
- **Snapshot crash.** Taking a snapshot of the graph view while another screen covered it crashed the native library; replaying the pictures off screen avoided the view altogether.

6 Testing and Evaluation

6.1 Evaluation method and controlling for bias

There were no test users. The evaluation therefore concentrates on the scoring metrics, as the proposal feedback advised: how they behave across different networks, and whether any factor dominates. Because the author built and evaluated the system, three guards against confirmation bias were used. First, properties and orderings were written and committed before the scoring formula; later versions extended them without weakening them. Second, property tests run on randomly generated networks rather than hand-picked ones [99]. Third, the sensitivity analysis is deterministic and regenerates its report from code, so it cannot be tuned by hand.

6.2 Correctness: unit and property-based tests

The suite has 284 tests in 25 files, all passing (Figure 9). Each of the 23 engine properties is checked over hundreds of random networks, for example: adding contact never lowers the score; time passing without data never raises it; a missed call changes nothing; and default ideals change nothing. Further tests cover layout, metrics, tie inference, encryption, erasure, call matching, trends, groups and plans (Appendix B).

```
✓ apps/mobile/test/health.test.ts (15 tests)
✓ apps/mobile/test/contacts.test.ts (12 tests)
✓ apps/mobile/test/arrange.test.ts (14 tests)
✓ apps/mobile/test/edit.test.ts (24 tests)
✓ apps/mobile/test/plans.test.ts (20 tests)
✓ apps/mobile/test/groups.test.ts (8 tests)
✓ apps/mobile/test/data.test.ts (12 tests)
✓ apps/mobile/test/metrics.test.ts (10 tests)
✓ apps/mobile/test/privacy.test.ts (17 tests)
✓ apps/mobile/test/interactions.test.ts (7 tests)
✓ packages/synth/test/orderings.test.ts (20 tests)
✓ packages/score-engine/test/explain.test.ts (8 tests)
✓ apps/mobile/test/graph-model.test.ts (8 tests)
✓ apps/mobile/test/person-stats.test.ts (4 tests)
✓ apps/mobile/test/ties.test.ts (9 tests)
✓ apps/mobile/test/live.test.ts (21 tests)
✓ apps/mobile/test/trends.test.ts (4 tests)
✓ apps/mobile/test/scenarios.test.ts (3 tests)
✓ apps/mobile/test/calendar.test.ts (4 tests)
✓ apps/mobile/test/report.test.ts (6 tests)
✓ packages/synth/test/analysis.test.ts (5 tests)
✓ apps/mobile/test/subset.test.ts (3 tests)
✓ apps/mobile/test/sample-data.test.ts (1 test)
✓ apps/mobile/test/layout.test.ts (24 tests)
✓ packages/score-engine/test/properties.test.ts (25 tests)
Test Files  25 passed (25)
     Tests  284 passed (284)
  Duration  1.34s (transform 1.04s, setup 0ms, collect 3.13s, tests 2.55s, environment 4ms, pre
```

Figure 9. The automated test suite: 284 tests in 25 files, all passing.

6.3 Behaviour on synthetic networks

Seventeen scenario orderings state which of two networks should score higher, written before the model. All hold. Table 6 shows representative scores.

Table 6. Engine totals (out of 100) for representative scenarios.

Scenario	Total	Expected relation
Close-knit, met in person / messages only	86.7 / 33.0	In person higher (O4)
Balanced six people / same contact with one person	79.5 / 19.4	Balanced higher (O2)
Large, important people neglected	54.8	Below close-knit (O1)
Busy period, paused / not paused	86.3 / 40.7	Paused higher (O8)
Bereavement, in memoriam / unmarked	86.3 / 66.9	Marked higher (O5)
Calls spread / all with one person	73.2 / 56.7	Spread higher (O12)
Calls only, nothing logged	49.9	Calls cannot replace catch-ups (O13)
Rated very good / strained	89.1 / 57.3	Good higher (O14)
Kept distant: ideal 3 / ideal 10 / ideal 3 with weekly meetings	75.2 / 61.6 / 61.6	Ideal respected (O16, O17)

Three results deserve comment. The gap between in-person and message-only contact (86.7 against 33.0) follows from the type weights, which rest on evidence that richer contact is associated with greater closeness and happiness [22], [100]; the exact weights are assumptions tested in Section 6.5. A network with calls but nothing logged scores 49.9: calls carry signal but cannot stand in for the user's own record, as intended by their 0.25 weight. And the ideal works in both directions: lowering it raised a deliberately distant relationship from 61.6 to 75.2, while weekly meetings despite the low ideal brought it back to 61.6.

6.4 Case studies: eight synthetic lives

To test the system on realistic patterns rather than isolated effects, eight lives were generated with fixed seeds and 300 days of history, and viewed in the app as of a fixed date (Table 7, Figure 10). People met in large numbers at work are logged as group catch-ups with unnamed counts.

Table 7. The eight synthetic lives.

Life	Score	5-month trend	What the app shows
Doctor	7	8 → 7	Patients as one group; a friend lost to work.
Kindergarten teacher	9	steady	A difficult parent at "Distant" matches a lowered ideal and costs nothing.
Waitress at a bar	7	6–8	Many contacts but loosely kept: the most catch-ups is not the closest network.
Trial lawyer	7	8 → 6 → 7	A long trial dents home and friends, then recovers.

Care-home resident	8	9 → 8	A friend's death, marked in memoriam, is not counted as neglect.
University student	8	dip at exams	A new friend made through a club; a home friend drifts.
Parent of three	9	flat	Daily contact with a teenager, yet "Drifting": quantity is not closeness.
New in town	8	9 → 6 → 8	The score dips after a move and recovers as new ties form.

The lives show the model separating quantity from quality (the waitress and the parent), respecting ideals (the teacher) and handling life events without blame (the care home, the move). They also expose a limitation: a busy professional with many unnamed contacts can still score well while losing friends, because the score averages across relationships.

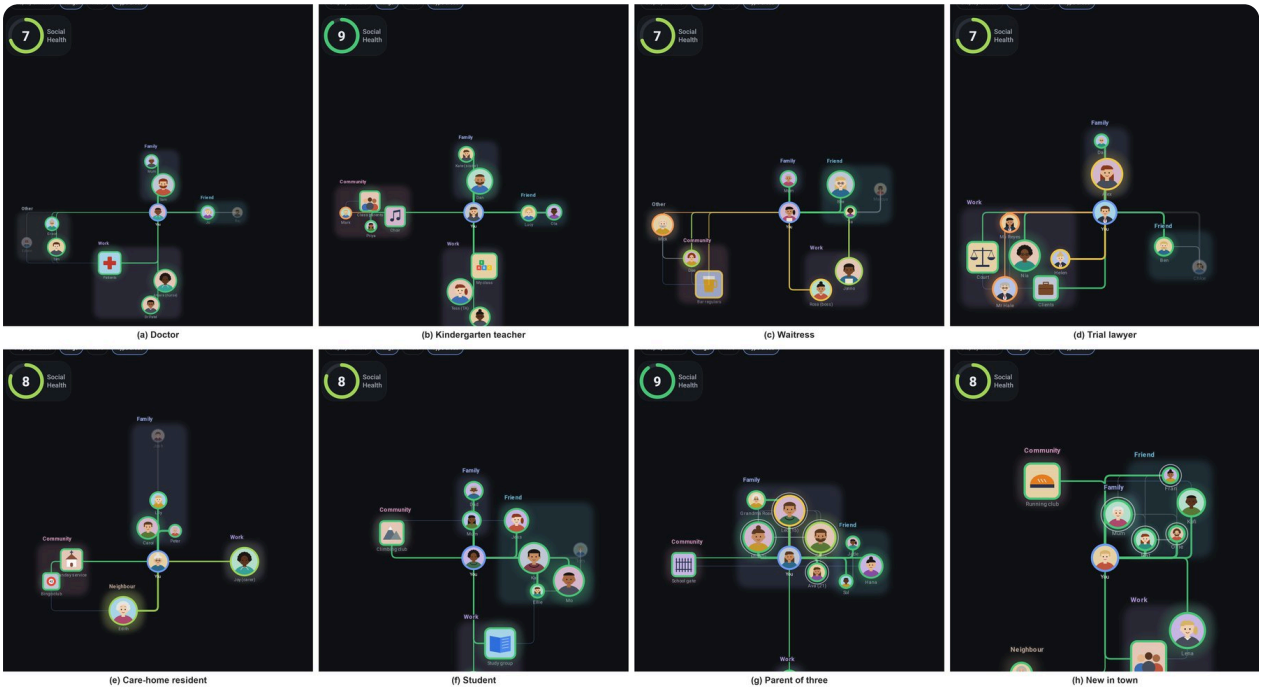


Figure 10. The eight synthetic lives, with illustrated avatars and group symbols.

The trends are as informative as the scores (Figure 11). The lawyer's score falls from 8 to 6 during an eighty-day trial and recovers to 7 afterwards, although catch-ups stayed high throughout. The newcomer's score dips from 9 to 6 after a move and recovers as new colleagues, a running club and a neighbour appear. In both cases the trend tells a story a user would recognise, which a static score would not.



Figure 11. Five-month trends for (a) the trial lawyer and (b) the newcomer.

6.5 Sensitivity, ablation and component dominance

Following composite-indicator practice [50], [51], each component was removed in turn, each weight halved and increased by half, and the main parameters varied. Agreement with the baseline ranking of the 23 scenarios was measured with Spearman's ρ [101].

- **Ablation:** removing any one component kept $\rho \geq 0.93$. Only one ordering broke: removing calls broke O15 (proper calls over brief ones), which is about calls by definition.
- **Weights:** every weight change kept $\rho \geq 0.98$ and broke no ordering; equal weights gave $\rho = 0.99$.
- **Parameters:** varying saturation, the recent-contact half-life, message weight and short-call credit kept $\rho \geq 0.96$ with no ordering broken.
- **Dominance:** over 500 random networks (Figure 12), the largest share of the total's variance attributable to one component was 0.33 (rhythm). Balance across types explained 0.26 despite a weight of 0.10, because it is derived from the same contact levels as rhythm.

The model is therefore robust to its weights, and no single factor dominates, but two components are correlated by construction.

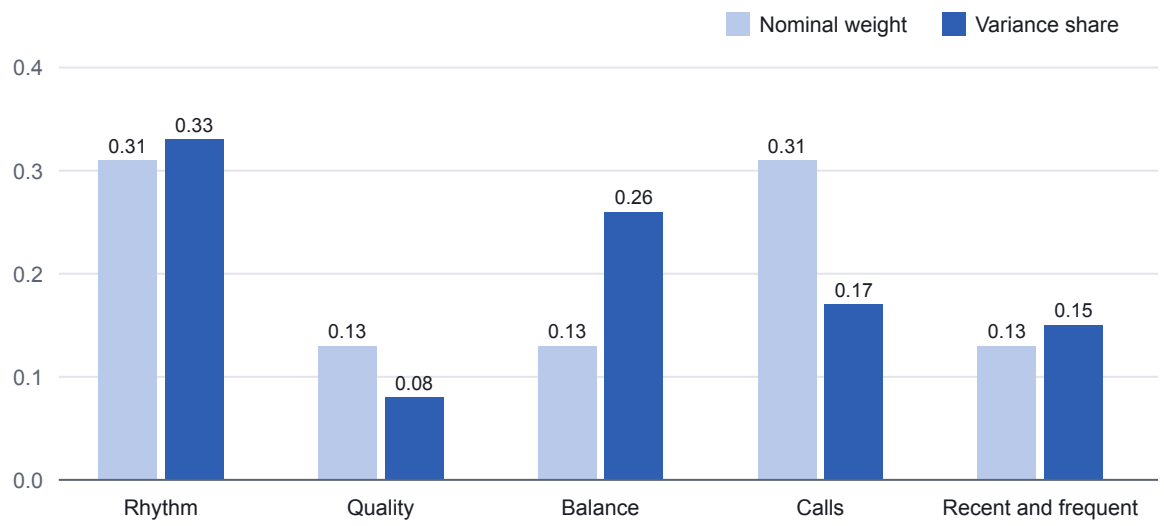


Figure 12. Weight (light) against share of variance (dark) per component over 500 random networks; the red line marks half.

Two caveats apply. Rank agreement is coarse, because the scenarios differ sharply; the largest change in a single total (28 points when short calls get no credit) shows that individual scores are more sensitive than rankings. And the random networks cover possible logs rather than real ones. Neither changes the conclusion that the weights could be adjusted, for example after a user study, without overturning the model's behaviour.

6.6 Performance on device

Table 8 summarises frame statistics recorded with the system's graphics profiler.

Table 8. Graph performance, release build on an OPPO CPH2765 phone, 150 synthetic people.

Measure	Result
Dropped frames while settling, at rest and interacting	≤ 0.5%
With about 250 links, settling	0.9–4.1% (27–34% before the batching fix)
Time to first frame	36–63 ms
Static drawing, Skia vs SVG	about 42 ms vs 110 ms
One-shot layout	about 1.3 s on the phone, 61 ms in Node

NFR5 is met. The bottleneck is layout computation in the Hermes engine, which the live layout hides by advancing one step per frame. The phone was an OPPO CPH2765 (Snapdragon SM8850, Android 16). The app was also exercised on a Samsung Galaxy Tab S9 Ultra (SM-X910) and an Android 16 tablet emulator (1600 × 2560 px) with no visible degradation.

6.7 Security and privacy verification

On the device, the app lock was checked by hand; an export made on the phone was decrypted on a laptop, a password one character off was rejected, and the file held no plaintext names; SQLCipher was confirmed active; and the release permissions were inspected. Unit tests cover encryption round trips, tampering and erasure.

6.8 Requirements traceability

Table 9 maps every requirement from Chapter 3 to where it is met and how it was verified.

Table 9. Requirements and how they were verified.

Requirement	Where met	Verified by
FR1–FR3	§4.2, §5.5	Edit, groups and plans tests; device use
FR4	§4.3	23 properties; 17 orderings
FR5	§4.4–4.5, §5.4	Metrics, ties and layout tests; eight lives
FR6	§5.5	Trend tests; screenshots
FR7	§4.6, §5.5	Call-matching tests; permission inspection
FR8	§5.5	Plan tests (one-off, repeating, shared)
FR9	§4.6, §5.3	Export round trip; erasure tests; report tests
NFR1–NFR2	§4.6	Threat model; device checks (§6.7)
NFR3–NFR4	§4.3, §6.5	Component display; sensitivity analysis
NFR5	§5.4	Frame statistics (§6.6)
NFR6	§3.4, §4.5	Design review against Table 3; not user-tested

7 Critical Reflection

7.1 Contribution against the research question

The project shows that a privacy-preserving, on-device application can turn everyday contact into an explainable measure of connectedness: network metrics, time-decayed scoring and a live visualisation all run on the phone, and every score can be explained component by component. The ideals answer the second part of the question: the model respects each user's hopes without changing anyone's score until they act. Whether the measure helps people is a different question, which this project could not answer.

7.2 Limitations and threats to validity

- **Not validated.** The weights are justified assumptions, shown to be robust but not validated against wellbeing outcomes.
- **Self-report.** Catch-ups are logged by hand and will be incomplete; the confidence measure flags sparse logging but cannot correct it.
- **Correlated components.** Balance across types shares its inputs with rhythm (§6.5).
- **Ideals relative to rhythm.** For a distant ideal, keeping strictly to a frequent rhythm also counts as closer than hoped; and the fit is symmetric, though users may mind distance more than closeness.
- **Averaging.** A high overall score can hide one important relationship fading; filtered scores and per-relationship bands mitigate this.
- **Synthetic evaluation.** Scenarios test whether the model does what its author intended, not whether that intention is right; they share a builder and a fixed date, and under-represent messy real logging.
- **Platform.** Call-log access is restricted on Google Play [44]; performance was measured on recent hardware only.

7.3 Ethical reflection

Recording other people, even privately, raises questions those people cannot answer. The design limits exposure: no server, no sharing, erasure and an anonymised export. Free-text notes, however, may say things about people that erasure cannot rewrite in catch-ups shared with others. The wellbeing mitigations in Table 3 were designed from the literature but not tested with users; a deployed version would need ethics review and a study watching for harm as well as benefit.

7.4 Process reflection

The proposal planned a server, reciprocal connections and a 0–100 score; the first two were dropped for privacy, and the score moved to 1–10 with descriptive bands after reflecting on the harms literature. Writing properties before formulas proved the most valuable decision: when ideals were added late, the properties showed at once what had changed. Several defects appeared only on real hardware, which argues for testing on devices early.

Some decisions would be made differently: balance across types should have used a different input from rhythm from the start; a small informal usability check would have tested the wording and colours the wellbeing requirement rests on; and the interaction model should have been prototyped on paper before late changes such as moving pages into panes.

8 Conclusion and Future Work

This project designed, built and evaluated a privacy-preserving personal social graph with an interpretable Social Health score. Requirements came from the literature, the five-component model was specified with testable properties before implementation, and the app keeps everything on the device. Evaluation without users showed the scoring correct against its specification, robust to its weights and not dominated by any component, with smooth rendering on real hardware.

Future work, in order of importance: a user study with ethics approval, measuring loneliness with validated scales before and after use [11], [12] and watching for harm; decorrelating the balance component from rhythm; a store-compatible source of call activity; importing encrypted exports; optional, gentle reminders; and moving layout computation off the main JavaScript thread.

References

[1] World Health Organization, *From Loneliness to Social Connection: Charting a Path to Healthier Societies. Report of the WHO Commission on Social Connection*. Geneva, Switzerland: WHO, 2025.

[2] World Health Organization, "Social connection linked to improved health and reduced risk of early death," News release, Jun. 30, 2025. [Online]. Available: https://www.who.int/news/item/30-06-2025-social-connection-linked-to-improved-heath-and-reduced-risk-of-early-death

[3] J. Holt-Lunstad, T. B. Smith, and J. B. Layton, "Social relationships and mortality risk: A meta-analytic review," *PLoS Medicine*, vol. 7, no. 7, Art. no. e1000316, 2010.

[4] J. Holt-Lunstad, T. B. Smith, M. Baker, T. Harris, and D. Stephenson, "Loneliness and social isolation as risk factors for mortality: A meta-analytic review," *Perspectives on Psychological Science*, vol. 10, no. 2, pp. 227–237, 2015.

[5] Office of the U.S. Surgeon General, *Our Epidemic of Loneliness and Isolation: The U.S. Surgeon General's Advisory on the Healing Effects of Social Connection and Community*. Washington, DC, USA: U.S. Department of Health and Human Services, 2023.

[6] B. J. Fogg, *Persuasive Technology: Using Computers to Change What We Think and Do*. San Francisco, CA, USA: Morgan Kaufmann, 2003.

[7] P. Verduyn, O. Ybarra, M. Résibois, J. Jonides, and E. Kross, "Do social network sites enhance or undermine subjective well-being? A critical review," *Social Issues and Policy Review*, vol. 11, no. 1, pp. 274–302, 2017.

[8] E. Kross *et al.*, "Facebook use predicts declines in subjective well-being in young adults," *PLoS ONE*, vol. 8, no. 8, Art. no. e69841, 2013.

[9] J. Holt-Lunstad, "Advancing social connection as a public health priority in the United States," *American Psychologist*, vol. 72, no. 6, pp. 517–530, 2017.

[10] D. Perlman and L. A. Peplau, "Toward a social psychology of loneliness," in *Personal Relationships 3: Personal Relationships in Disorder*, S. Duck and R. Gilmour, Eds. London, U.K.: Academic Press, 1981, pp. 31–56.

[11] D. W. Russell, "UCLA Loneliness Scale (Version 3): Reliability, validity, and factor structure," *Journal of Personality Assessment*, vol. 66, no. 1, pp. 20–40, 1996.

[12] M. E. Hughes, L. J. Waite, L. C. Hawkley, and J. T. Cacioppo, "A short scale for measuring loneliness in large surveys: Results from two population-based studies," *Research on Aging*, vol. 26, no. 6, pp. 655–672, 2004.

[13] J. de Jong Gierveld and T. van Tilburg, "A 6-item scale for overall, emotional, and social loneliness: Confirmatory tests on survey data," *Research on Aging*, vol. 28, no. 5, pp. 582–598, 2006, doi: 10.1177/0164027506289723.

[14] J. Lubben, E. Blozik, G. Gillmann, S. Iliffe, W. von Renteln Kruse, J. C. Beck, and A. E. Stuck, "Performance of an abbreviated version of the Lubben Social Network Scale among three European community-dwelling older adult populations," *The Gerontologist*, vol. 46, no. 4, pp. 503–513, 2006.

[15] Office for National Statistics, "Measuring loneliness: Guidance for use of the national indicators on surveys," ONS, Newport, U.K., Dec. 2018.

[16] T. Hansen, R. B. Nes, K. Hynek, T. S. Nilsen, A. Reneflot, K. Stene-Larsen, R. A. Tornes, and J. Bidonde, "Tackling social disconnection: An umbrella review of RCT-based interventions targeting social isolation and loneliness," *BMC Public Health*, vol. 24, 2024, doi: 10.1186/s12889-024-19396-8.

[17] B. L. Perry, B. A. Pescosolido, and S. P. Borgatti, *Egocentric Network Analysis: Foundations, Methods, and Models*. Cambridge, U.K.: Cambridge Univ. Press, 2018.

[18] R. I. M. Dunbar, "Neocortex size as a constraint on group size in primates," *Journal of Human Evolution*, vol. 22, no. 6, pp. 469–493, 1992.

[19] R. A. Hill and R. I. M. Dunbar, "Social network size in humans," *Human Nature*, vol. 14, no. 1, pp. 53–72, 2003.

[20] W.-X. Zhou, D. Sornette, R. A. Hill, and R. I. M. Dunbar, "Discrete hierarchical organization of social group sizes," *Proceedings of the Royal Society B*, vol. 272, no. 1561, pp. 439–444, 2005.

[21] J. A. Hall, "How many hours does it take to make a friend?," *Journal of Social and Personal Relationships*, vol. 36, no. 4, pp. 1278–1296, 2019.

[22] S. G. B. Roberts and R. I. M. Dunbar, "The costs of family and friends: An 18-month longitudinal study of relationship maintenance and decay," *Evolution and Human Behavior*, vol. 32, no. 3, pp. 186–197, 2011.

[23] R. S. Burt, "Decay functions," *Social Networks*, vol. 22, no. 1, pp. 1–28, 2000.

[24] M. S. Granovetter, "The strength of weak ties," *American Journal of Sociology*, vol. 78, no. 6, pp. 1360–1380, 1973.

[25] P. V. Marsden and K. E. Campbell, "Measuring tie strength," *Social Forces*, vol. 63, no. 2, pp. 482–501, 1984.

[26] T. Opsahl, F. Agneessens, and J. Skvoretz, "Node centrality in weighted networks: Generalizing degree and shortest paths," *Social Networks*, vol. 32, no. 3, pp. 245–251, 2010.

[27] D. J. Watts and S. H. Strogatz, "Collective dynamics of 'small-world' networks," *Nature*, vol. 393, pp. 440–442, 1998.

[28] L. C. Freeman, "A set of measures of centrality based on betweenness," *Sociometry*, vol. 40, no. 1, pp. 35–41, 1977.

[29] U. Brandes, "A faster algorithm for betweenness centrality," *Journal of Mathematical Sociology*, vol. 25, no. 2, pp. 163–177, 2001.

[30] R. S. Burt, *Structural Holes: The Social Structure of Competition*. Cambridge, MA, USA: Harvard Univ. Press, 1992.

[31] S. P. Borgatti, "Structural holes: Unpacking Burt's redundancy measures," *Connections*, vol. 20, no. 1, pp. 35–38, 1997.

[32] M. E. J. Newman, "Modularity and community structure in networks," *Proceedings of the National Academy of Sciences*, vol. 103, no. 23, pp. 8577–8582, 2006.

[33] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of Statistical Mechanics: Theory and Experiment*, Art. no. P10008, 2008.

[34] N. Eagle, A. Pentland, and D. Lazer, "Inferring friendship network structure by using mobile phone data," *Proceedings of the National Academy of Sciences*, vol. 106, no. 36, pp. 15274–15278, 2009.

[35] J.-P. Onnela, J. Saramäki, J. Hyvönen, G. Szabó, D. Lazer, K. Kaski, J. Kertész, and A.-L. Barabási, "Structure and tie strengths in mobile communication networks," *Proceedings of the National Academy of Sciences*, vol. 104, no. 18, pp. 7332–7336, 2007.

[36] M. M. Qirtas, E. Zafeiridi, D. Pesch, and E. Bantry White, "Loneliness and social isolation detection using passive sensing techniques: Scoping review," *JMIR mHealth and uHealth*, vol. 10, no. 4, Art. no. e34638, 2022, doi: 10.2196/34638.

[37] A. Doryab *et al.*, "Identifying behavioral phenotypes of loneliness and social isolation with passive sensing: Statistical analysis, data mining and machine learning of smartphone and Fitbit data," *JMIR mHealth and uHealth*, vol. 7, no. 7, Art. no. e13209, 2019.

[38] S. Saeb, M. Zhang, C. J. Karr, S. M. Schueller, M. E. Corden, K. P. Kording, and D. C. Mohr, "Mobile phone sensor correlates of depressive symptom severity in daily-life behavior: An exploratory study," *Journal of Medical Internet Research*, vol. 17, no. 7, Art. no. e175, 2015, doi: 10.2196/jmir.4273.

[39] Z. Li, D. Shi, F. Wang, and F. Liu, "Loneliness recognition based on mobile phone data," in *Proc. 2016 Int. Symp. Advances in Electrical, Electronics and Computer Engineering (ISAEECE)*, Guangzhou, China, 2016. Paris, France: Atlantis Press. [Online]. Available: https://www.atlantis-press.com/proceedings/isaeece-16/25852860

[40] Z. Yang *et al.*, "Loneliness forecasting using multi-modal wearable and mobile sensing in everyday settings," 2024, *arXiv:2410.00020*.

[41] S. Jafarlou *et al.*, "Objective monitoring of loneliness levels using smart devices: A multi-device approach for mental health applications," *PLoS ONE*, vol. 19, no. 6, Art. no. e0298949, 2024, doi: 10.1371/journal.pone.0298949.

[42] J. Mayer, P. Mutchler, and J. C. Mitchell, "Evaluating the privacy properties of telephone metadata," *Proceedings of the National Academy of Sciences*, vol. 113, no. 20, pp. 5536–5541, 2016.

[43] Y.-A. de Montjoye, C. A. Hidalgo, M. Verleysen, and V. D. Blondel, "Unique in the crowd: The privacy bounds of human mobility," *Scientific Reports*, vol. 3, Art. no. 1376, 2013.

[44] Google, "Use of SMS or Call Log permission groups," Play Console Help. [Online]. Available: https://support.google.com/googleplay/android-developer/answer/10208820 (accessed Oct. 8, 2026).

[45] Z. Bar-Yossef and N. Kraus, "Context-sensitive query auto-completion," in *Proc. 20th Int. Conf. World Wide Web (WWW)*, Hyderabad, India, 2011, pp. 107–116.

[46] M. Shokouhi and K. Radinsky, "Time-sensitive query auto-completion," in *Proc. 35th Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, Portland, OR, USA, 2012, pp. 601–610.

[47] S. Whiting and J. M. Jose, "Recent and robust query auto-completion," in *Proc. 23rd Int. Conf. World Wide Web (WWW)*, Seoul, South Korea, 2014, pp. 971–982.

[48] Mozilla, "Ranking," Firefox Source Docs: Address Bar. [Online]. Available: https://firefox-source-docs.mozilla.org/browser/urlbar/ranking.html (accessed Oct. 12, 2026).

[49] G. Cormode, V. Shkapenyuk, D. Srivastava, and B. Xu, "Forward decay: A practical time decay model for streaming systems," in *Proc. IEEE 25th Int. Conf. Data Engineering (ICDE)*, Shanghai, China, 2009, pp. 138–149.

[50] OECD and European Commission Joint Research Centre, *Handbook on Constructing Composite Indicators: Methodology and User Guide*. Paris, France: OECD Publishing, 2008.

[51] A. Saltelli *et al.*, *Global Sensitivity Analysis: The Primer*. Chichester, U.K.: Wiley, 2008.

[52] Monica, "Monica: Personal relationship manager." [Online]. Available: https://monicahq.com (accessed Oct. 12, 2026).

[53] M. Burke and R. E. Kraut, "The relationship between Facebook use and well-being depends on communication type and tie strength," *Journal of Computer-Mediated Communication*, vol. 21, no. 4, pp. 265–281, 2016.

[54] S. G. S. Shah, D. Nogueras, H. C. van Woerden, and V. Kiparoglou, "Evaluation of the effectiveness of digital technology interventions to reduce loneliness in older adults: Systematic review and meta-analysis," *Journal of Medical Internet Research*, vol. 23, no. 6, Art. no. e24712, 2021, doi: 10.2196/24712.

[55] L. Harkin, C. Stevenson, E. Fino, C. Talbot, A. Stuart, M. S.-A. Park, J. Lay, D. Katz, B. Price, and D. Gooch, "Older adults can re-appraise loneliness using a social connectivity app: A mixed method intervention study," *Aging & Mental Health*, vol. 30, no. 3, pp. 517–528, 2026, doi: 10.1080/13607863.2025.2572017.

[56] I. Li, A. Dey, and J. Forlizzi, "A stage-based model of personal informatics systems," in *Proc. SIGCHI Conf. Human Factors in Computing Systems (CHI)*, Atlanta, GA, USA, 2010, pp. 557–566.

[57] D. A. Epstein, A. Ping, J. Fogarty, and S. A. Munson, "A lived informatics model of personal informatics," in *Proc. ACM Int. Joint Conf. Pervasive and Ubiquitous Computing (UbiComp)*, Osaka, Japan, 2015, pp. 731–742.

[58] D. Lupton, *The Quantified Self: A Sociology of Self-Tracking*. Cambridge, U.K.: Polity Press, 2016.

[59] J. Etkin, "The hidden cost of personal quantification," *Journal of Consumer Research*, vol. 42, no. 6, pp. 967–984, 2016.

[60] R. L. Kahn and T. C. Antonucci, "Convoys over the life course: Attachment, roles, and social support," in *Life-Span Development and Behavior*, vol. 3, P. B. Baltes and O. G. Brim, Eds. New York, NY, USA: Academic Press, 1980, pp. 253–286.

[61] B. Hogan, J. A. Carrasco, and B. Wellman, "Visualizing personal networks: Working with participant-aided sociograms," *Field Methods*, vol. 19, no. 2, pp. 116–144, 2007.

[62] T. M. J. Fruchterman and E. M. Reingold, "Graph drawing by force-directed placement," *Software: Practice and Experience*, vol. 21, no. 11, pp. 1129–1164, 1991.

[63] M. Bostock, V. Ogievetsky, and J. Heer, "D³: Data-driven documents," *IEEE Transactions on Visualization and Computer Graphics*, vol. 17, no. 12, pp. 2301–2309, 2011.

[64] W. S. Cleveland and R. McGill, "Graphical perception: Theory, experimentation, and application to the development of graphical methods," *Journal of the American Statistical Association*, vol. 79, no. 387, pp. 531–554, 1984.

[65] J. Mackinlay, "Automating the design of graphical presentations of relational information," *ACM Transactions on Graphics*, vol. 5, no. 2, pp. 110–141, 1986, doi: 10.1145/22949.22950.

[66] B. Alsallakh, L. Micallef, W. Aigner, H. Hauser, S. Miksch, and P. Rodgers, "The state-of-the-art of set visualization," *Computer Graphics Forum*, vol. 35, no. 1, pp. 234–260, 2016.

[67] C. Collins, G. Penn, and S. Carpendale, "Bubble Sets: Revealing set relations with isocontours over existing visualizations," *IEEE Transactions on Visualization and Computer Graphics*, vol. 15, no. 6, pp. 1009–1016, 2009.

[68] "Regulation (EU) 2016/679 (UK GDPR), Article 5: Principles relating to processing of personal data," legislation.gov.uk. [Online]. Available: https://www.legislation.gov.uk/eur/2016/679/article/5 (accessed Oct. 12, 2026).

[69] European Parliament and Council of the European Union, "Regulation (EU) 2016/679 (General Data Protection Regulation)," *Official Journal of the European Union*, vol. L119, pp. 1–88, May 2016.

[70] Information Commissioner's Office, "Exemptions," UK GDPR guidance and resources. [Online]. Available: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/exemptions/ (accessed Oct. 12, 2026).

[71] A. Cavoukian, "Privacy by Design: The 7 foundational principles," Information and Privacy Commissioner of Ontario, Toronto, ON, Canada, 2009.

[72] M. Kleppmann, A. Wiggins, P. van Hardenberg, and M. McGranaghan, "Local-first software: You own your data, in spite of the cloud," in *Proc. ACM SIGPLAN Int. Symp. New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!)*, Athens, Greece, 2019, pp. 154–178.

[73] Android Developers, "Android Keystore system." [Online]. Available: https://developer.android.com/privacy-and-security/keystore (accessed Oct. 8, 2026).

[74] Android Developers, "Back up user data with Auto Backup." [Online]. Available: https://developer.android.com/identity/data/autobackup (accessed Oct. 12, 2026).

[75] OWASP Foundation, "OWASP Mobile Application Security Verification Standard (MASVS)." [Online]. Available: https://mas.owasp.org/MASVS/ (accessed Oct. 12, 2026).

[76] C. Wrzus, M. Hänel, J. Wagner, and F. J. Neyer, "Social network changes and life events across the life span: A meta-analysis," *Psychological Bulletin*, vol. 139, no. 1, pp. 53–80, 2013, doi: 10.1037/a0028601.

[77] G. E. Krasner and S. T. Pope, "A cookbook for using the model-view controller user interface paradigm in Smalltalk-80," *Journal of Object-Oriented Programming*, vol. 1, no. 3, pp. 26–49, 1988.

[78] M. Fowler, "Event sourcing," martinfowler.com, Dec. 2005. [Online]. Available: https://martinfowler.com/eaaDev/EventSourcing.html (accessed Oct. 8, 2026).

[79] M. E. J. Newman, "Scientific collaboration networks. II. Shortest paths, weighted networks, and centrality," *Physical Review E*, vol. 64, no. 1, Art. no. 016132, 2001.

[80] A. Shostack, *Threat Modeling: Designing for Security*. Indianapolis, IN, USA: Wiley, 2014.

[81] Zetetic LLC, "SQLCipher: Full database encryption for SQLite." [Online]. Available: https://www.zetetic.net/sqlcipher/ (accessed Oct. 8, 2026).

[82] M. Dworkin, "Recommendation for block cipher modes of operation: Galois/Counter Mode (GCM) and GMAC," NIST, Gaithersburg, MD, USA, Special Publication 800-38D, 2007.

[83] M. S. Turan, E. Barker, W. Burr, and L. Chen, "Recommendation for password-based key derivation, Part 1: Storage applications," NIST, Gaithersburg, MD, USA, Special Publication 800-132, 2010.

[84] OWASP Foundation, "Password storage cheat sheet," OWASP Cheat Sheet Series. [Online]. Available: https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html (accessed Oct. 7, 2026).

[85] Android Developers, "KeyGenParameterSpec.Builder: setInvalidatedByBiometricEnrollment." [Online]. Available: https://developer.android.com/reference/android/security/keystore/KeyGenParameterSpec.Builder (accessed Oct. 12, 2026).

[86] Meta Platforms, "React Native." [Online]. Available: https://reactnative.dev (accessed Oct. 8, 2026).

[87] Expo, "Expo documentation." [Online]. Available: https://docs.expo.dev (accessed Oct. 8, 2026).

[88] Microsoft, "TypeScript." [Online]. Available: https://www.typescriptlang.org (accessed Oct. 8, 2026).

[89] Shopify, "React Native Skia." [Online]. Available: https://shopify.github.io/react-native-skia/ (accessed Oct. 8, 2026).

[90] Google, "Skia graphics library." [Online]. Available: https://skia.org (accessed Oct. 8, 2026).

[91] Software Mansion, "React Native Reanimated." [Online]. Available: https://docs.swmansion.com/react-native-reanimated/ (accessed Oct. 8, 2026).

[92] Software Mansion, "React Native Gesture Handler." [Online]. Available: https://docs.swmansion.com/react-native-gesture-handler/ (accessed Oct. 8, 2026).

[93] M. Bostock, "d3-force: Force-directed graph layout using velocity Verlet integration." [Online]. Available: https://d3js.org/d3-force (accessed Oct. 8, 2026).

[94] Expo, "SQLite," Expo SDK documentation. [Online]. Available: https://docs.expo.dev/versions/latest/sdk/sqlite/ (accessed Oct. 12, 2026).

[95] Margelo, "react-native-quick-crypto." [Online]. Available: https://github.com/margelo/react-native-quick-crypto (accessed Oct. 8, 2026).

[96] Vitest, "Vitest: Next generation testing framework." [Online]. Available: https://vitest.dev (accessed Oct. 8, 2026).

[97] N. Dubien, "fast-check: Property based testing framework for JavaScript and TypeScript." [Online]. Available: h ttps://fast-check.dev (accessed Oct. 8, 2026).

[98] Meta Platforms, "Hermes JavaScript engine." [Online]. Available: https://hermesengine.dev (accessed Oct. 8, 2026).

[99] K. Claessen and J. Hughes, "QuickCheck: A lightweight tool for random testing of Haskell programs," in *Proc. 5th ACM SIGPLAN Int. Conf. Functional Programming (ICFP)*, Montreal, Canada, 2000, pp. 268–279.

[100] T. A. Vlahovic, S. Roberts, and R. Dunbar, "Effects of duration and laughter on subjective happiness within different modes of communication," *Journal of Computer-Mediated Communication*, vol. 17, no. 4, pp. 436–450, 2012.

[101] C. Spearman, "The proof and measurement of association between two things," *The American Journal of Psychology*, vol. 15, no. 1, pp. 72–101, 1904.

Appendix A: Scoring model parameters

Parameter	Value
Type weights	in person 1, video 0.7, voice 0.5, messages 0.2
Quality multiplier (unrated; ratings 1–5)	1; 0, 0.4, 0.75, 0.9, 1
Default rhythm	at least once a month (30 days); long-distance doubles the interval
Saturation constant k	1
Together decay rate	0.1
Recent-and-frequent half-life	30 days
Short calls (< 60 s), missed calls	0.3, 0
Lost touch	no contact for more than 3 × the interval and at least 30 days
Tie half-life; minimum tie weight	90 days; 0.2

Appendix B: Property and ordering catalogue

Properties. P1 bounded · P2 deterministic · P3 adding contact never lowers the score · P4 time passing without new data never raises it · P5 diminishing returns per person · P6 invariant to relabelling and record order · P7 moving contact to a neglected, equally important person raises rhythm · P8 contact with excluded relationships never raises it · P9 adding a pause never lowers it · P10 empty network scores 0 · P11 an answered call never lowers it · P12 a missed call changes nothing · P13 diminishing returns for calls · P14 an unconnected call source is left out · P15 recent-and-frequent contact cannot be inflated through one person · P16 frequent recent contact outranks one old contact · P17 default ideals change nothing · P18 "don't mind" never changes the score · P19 fit is 1 at the ideal and falls away from it · P20 much contact with a relationship hoped to be distant eventually lowers the score · P21 a custom frequency equal to a preset scores like it · P22 marking someone as together never lowers the score · P23 marking no one as together changes nothing.

Orderings. O1 close-knit on rhythm > large but neglected · O2 balanced > concentrated · O3 active > inactive · O4 in person > messages only · O5 in memoriam removes the neglect penalty · O6 batch logging lowers confidence · O7 harmful contact does not help · O8 paused gap > unpaused gap · O9 single type not heavily penalised · O10 healthy ≥ 60, neglected ≤ 40 · O11 regular calls > no calls · O12 calls spread > calls with one person · O13 calls alone cannot replace catch-ups · O14 good relationships > strained · O15 proper calls > brief calls · O16 a distant ideal is respected · O17 with a distant ideal, sparse contact > weekly closeness.

Appendix C: Threat model summary

Threat	Mitigation	Residual risk
Someone opens the unlocked app	App lock on by default	Anyone who knows the PIN
Storage read from a stolen phone	SQLCipher, Keystore key, sandbox	Rooted device (out of scope)
Cloud backup copies data	Backup disabled	Vendor transfer tools
Export read by others	AES-256-GCM, PBKDF2 600,000	Weak passwords
Deleted records recovered	Secure delete; key destruction	Flash wear-levelling (encrypted)
Address book or call log exposed	On-demand permissions; names only; no numbers	Access while granted
Distress from the score	Descriptive wording; hide option; exclusions	Untested with users

Appendix D: Scenario details

The scenario data, screenshots and descriptions are in the project repository under `docs/report/scenarios`, and can be reproduced in the app from the engine preview.